OpenVLA · arXiv 2406.09246 · CoRL 2024

Scientific Reviewer

8+3

Open-science strengths + comparison caveats
Strong CoRL’24 systems paper — not 2026 substrate

Oct 21 paper club · VLA Architectures

Spine: naïve discrete AR @ ~5–6 Hz · historical open AR baseline

BEHAVIOR deploy = $\pi_{0.5}$ / OpenPI / GR00T · OpenVLA = open AR control

Action-family taxonomy · CLUB SPINE

Must appear in every role · no-chunk AR ≈ 5–6 Hz

Family	Model	Signature
Naïve discrete AR bins	**OpenVLA / RT-2**	Lang grounding @ ~5–6 Hz; weak high-Hz
DDPM chunks	**Octo / DP**	Smoother narrow skills
FM chunks	**π_0 / $\pi_{0.5}$**	2025–26 deploy winners
Compressed discrete AR	**FAST / OpenVLA+FAST / π_0-FAST**	AR recovers high-Hz

Syllabus: OpenVLA = historical open AR baseline · BEHAVIOR deploy = $\pi_{0.5}$ / OpenPI / GR00T

Reviewer strengths · steelman

Open 7B + OXE

First widely usable open VLA with real multi-embodiment OOB + full PyTorch/HF stack

PEFT / quant tables

LoRA + int4 are the adoption unlock — Empiricist-critical, auditable VRAM/SR

Ablations actionable

OpenX mix (−30), fused vision (−5), vision FT, quantization latency

Honest limits

Single cam, ~6 Hz, no chunks, <90% reliability — Discussion owns the gaps

Internet VLM → robot

Beats much larger closed RT-2-X on many axes with curated OpenX data

Reviewer weaknesses / caveats

AR single-step = 2024 local optimum

By 2025–26 FM+chunks dominate — cite as open stack, not final control prior

+16.5 pp vs RT-2-X not architecture-only

Embodiment suites · 970k vs 350k · closed API · App C 2nd-token hack

+20.4% vs DP is conditional

Lang multi-object win; DP competitive/better on narrow dexterity & LIBERO-Object

LIBERO margins thin

76.5 vs 75.1 vs 72.4 — rank story > raw SR; real→sim gap

Crude 256 bins · single RGB

FAST later diagnoses naïve binning; no wrist/proprio/history

Strong CoRL'24 open-science systems paper

Cite for: open 7B VLA + LoRA/int4 · VLM→discrete actions · lang FT

Do NOT cite for: 2026 BEHAVIOR SOTA · best high-Hz dexterity · chunked FM replacement

Scientific core (open Prismatic AR VLA + OXE + LoRA) holds.

Do not inflate to “solved generalist manipulation” or “2026 substrate.”

What to cite OpenVLA for (2026)

CITE FOR

- ✓ Open 7B VLA + LoRA/int4 recipes
- ✓ VLM backbone → discrete actions works
- ✓ Language grounding FT default
- ✓ Ablation morals: mix, vision FT, fused vision
- ✓ Historical open AR baseline in syllabus

DO NOT CITE FOR

- ✗ 2026 BEHAVIOR SOTA control
- ✗ Best high-Hz dexterity
- ✗ Replacement for chunked FM/DP
- ✗ End-to-end multi-cam mobile bimanual
- ✗ Solved generalist manipulation

Novelty is open VLA systems + PEFT/quant — not a new control law

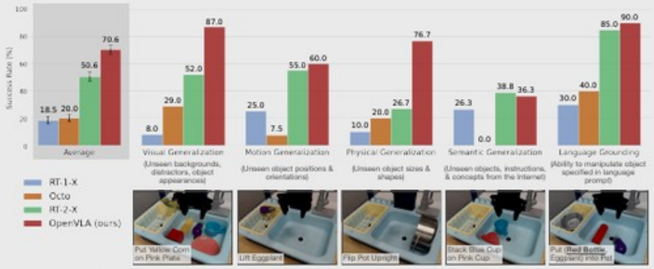

Figure 3: **BridgeData V2 WidowX robot evaluation tasks and results.** We evaluate OpenVLA and prior state-of-the-art generalist robot policies on a comprehensive suite of tasks covering several axes of generalization, as well as tasks that specifically assess language conditioning ability. OpenVLA achieves highest overall performance and even outperforms closed-source model RT-2-X in all categories except for semantic generalization. Average success rates ± StdErr are computed across 170 total rollouts per approach. See Table 4 for detailed results.

techniques for large transformer model training such as automatic mixed precision (AMP, PyTorch [75]), FlashAttention [76], and fully sharded data parallelism (FSDP, Zhao et al. [77]). Out of the box, the OpenVLA codebase has full support for training on the Open X dataset, integrates with HuggingFace's [21] `AutoModel` class, and supports LoRA fine-tuning [26] and quantized model inference [27, 88].

5 Experiments

The goal of our experimental evaluations is to test OpenVLA's ability to serve as a powerful multi-robot control policy out of the box, as well as be a good initialization for fine-tuning to new robot tasks. Concretely, we aim to answer the following questions:

1. How does OpenVLA compare to prior generalist robot policies, when evaluating on multiple robots and various types of generalization?
2. Can OpenVLA be effectively fine-tuned on a new robot setup and task, and how does it compare to state-of-the-art data-efficient imitation learning approaches?
3. Can we use parameter-efficient fine-tuning and quantization to reduce the computational requirements for training and inference of OpenVLA models and make them more accessible? What are the performance-compute trade-offs?

5.1 Direct Evaluations on Multiple Robot Platforms

Robot Setups and Tasks. We evaluate OpenVLA's performance "out-of-the-box" on two robot embodiments: the WidowX robot from the BridgeData V2 evaluations [6] (see Fig. 1, left) and the mobile manipulation robot from the RT-1 and RT-2 evaluations [2, 7] ("Google robot"; see Fig. 1, middle). Both platforms have been extensively used in prior works for evaluating generalist robot policies [1, 2, 5, 7]. We define a comprehensive set of evaluation tasks in each environment that covers various axes of generalization, such as **visual** (unseen backgrounds, distractor objects, colors/appearances of objects); **motion** (unseen object positions/orientations); **physical** (unseen object sizes/shapes); and **semantic** (unseen target objects, instructions, and concepts from the Internet) generalization. We also assess language conditioning ability in scenes with multiple objects, testing whether the policy can manipulate the correct target object, as specified in the user's prompt. See bottom row of Fig. 3 and Fig. 4 for example task images in the BridgeData V2 and Google robot evaluations, respectively. Overall, we evaluated each method in 170 rollouts (17 tasks with 10 trials each) for BridgeData V2 experiments and 60 rollouts (12 tasks with 5 trials each) for Google robot experiments. A detailed breakdown of all tasks and how they differ from the training data is in

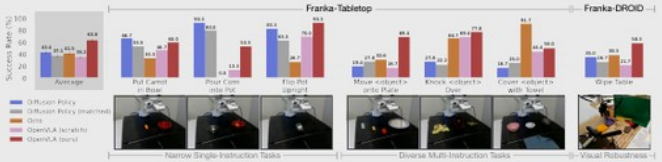

Figure 5: **Adapting to new robot setups.** We evaluate the state-of-the-art Diffusion Policy trained from scratch on seven Franka Emika Panda tasks (10–150 demonstrations each), as well as generalist robot policies Octo and OpenVLA fine-tuned on the same data. Diffusion Policy exhibits strong performance on narrow single-instruction tasks, while Octo and OpenVLA perform better on diverse fine-tuning tasks involving multiple instructions and distractor objects. Overall, OpenVLA achieves highest aggregate performance across both setups, suggesting that it is an effective default for learning a policy on a downstream task. Average success rates $\pm$ StdErr are computed across 129 rollouts per approach (99 for Franka-Tabletop tasks and 30 for Franka-DROID tasks). See Table 7 for detailed results.

mounted on a movable standing desk. The setups use 5Hz and 15 Hz non-blocking controllers, respectively. We choose Franka robot arms as the target embodiment for our fine-tuning experiments since they are widely used in the robot learning community and thus a likely "target" of OpenVLA fine-tuning. We test on setups with different control frequencies to test OpenVLA's applicability to a range of use cases.

Comparisons. We compare to **Diffusion Policy** [3], a state-of-the-art data-efficient imitation learning approach, trained from scratch. We also compare to **Diffusion Policy (matched)**, a version of Diffusion Policy that matches the input and output specifications of OpenVLA.[3] Additionally, we evaluate **Octo** [5] fine-tuned on the target dataset, since it is currently the best generalist policy that supports fine-tuning (fine-tuning of RT-2-X is not supported through its inference API). We also fine-tune OpenVLA on the same target dataset, and the resulting policy is denoted by **OpenVLA**. Finally, as an ablation experiment, we compare to **OpenVLA (scratch)**, where we directly fine-tune the underlying base Prismatic VLM on the target robot setup – rather than fine-tuning the OpenX-pretrained OpenVLA model – to assess the benefit of large-scale robot pretraining.

We present the results in Fig. 5 (per-task breakdown in Appendix, Table 7). We find that both versions of Diffusion Policy are competitive with or outperform the generalist policies Octo and OpenVLA on narrower single-instruction tasks like "Put Carrot in Bowl" and "Pour Corn into Pot", but the pretrained generalist policies perform better in more diverse fine-tuning tasks that involve multiple objects in the scene and require language conditioning. OpenX pretraining for Octo and OpenVLA enables the models to better adapt to these more diverse tasks where language grounding is important; we see evidence for this in the lower performance of OpenVLA (scratch).

Overall, we find that OpenVLA achieves the highest average performance. Notably, most prior works achieve strong performance only in *either* narrow single-instruction *or* diverse multi-instruction tasks, resulting in widely varying success rates. OpenVLA is the only approach that achieves at least 50% success rate across all tested tasks, suggesting that it can be a strong default option for imitation learning tasks, particularly if they involve a diverse set of language instructions. For narrower but highly dexterous tasks, Diffusion Policy still shows smoother and more precise trajectories; incorporating action chunking and temporal smoothing, as implemented in Diffusion Policy, may help OpenVLA attain the same level of dexterity and may be a promising direction for future work (see Section 6 for a detailed discussion of current limitations).

[3] The full Diffusion Policy uses a two-step observation history with both images and proprioceptive state, and performs receding horizon control by predicting a chunk of T future actions and executing the first X actions in open-loop fashion before predicting the next chunk (for 15Hz control, we set $T = 16, X = 8$ like in the DROID prior work [11]; for 5Hz control, we reduce the chunk sizes to $T = 8, X = 3$). It is also the only method in Section 5.2 that predicts *absolute* Cartesian coordinates to control the robot; all other methods use *relative* position control. Diffusion Policy (matched) uses a single image as input, has no proprioceptive information and no observation history, and predicts a single relative position control action without action chunking.

LIBERO sim · Table 12 · LoRA FT

Method	Spatial	Object	Goal	Long	Avg / Rank
DP scratch	78.3	92.5 ★	68.3	50.5	72.4 / 2.5
Octo FT	78.9	85.7	84.6 ★	51.1	75.1 / 2
OpenVLA LoRA	84.7 ★	88.4	79.2	53.7 ★	76.5 / 1.5

Pattern: OpenVLA wins Spatial/Long avg rank · DP wins Object · Octo wins Goal

Hygiene: drop no-ops · filter fails · 180° rotate · third-person · regen 256→224

Thin margins — do not treat as decisive VLA superiority

Scientific Reviewer · close

Cite as open 7B VLA + PEFT/quant — not 2026 substrate.

Caveats: RT-2-X quirks · DP wins narrow dexterity · thin LIBERO deltas

Q&A · Oct 21 · OpenVLA / 2406.09246

OpenVLA: An Open-Source Vision-Language-Action Model

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https://openvla.github.io

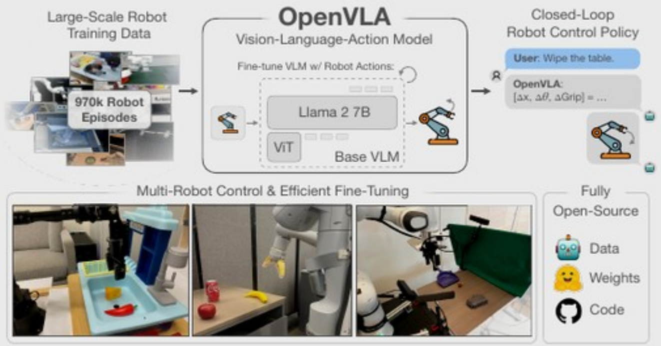

Figure 1: We present OpenVLA, a 7B-parameter open-source vision-language-action model (VLA), trained on 970k robot episodes from the Open X-Embodiment dataset [1]. OpenVLA sets a new state of the art for generalist robot manipulation policies. It supports controlling multiple robots out of the box and can be quickly adapted to new robot domains via parameter-efficient fine-tuning. The OpenVLA checkpoints and PyTorch training pipeline are fully open-source and models can be downloaded and fine-tuned from HuggingFace.

Abstract: Large policies pretrained on a combination of Internet-scale vision-language data and diverse robot demonstrations have the potential to change how we teach robots new skills: rather than training new behaviors from scratch, we can fine-tune such vision-language-action (VLA) models to obtain robust, generalizable policies for visuomotor control. Yet, widespread adoption of VLAs for robotics has been challenging as 1) existing VLAs are largely closed and inaccessible to the public, and 2) prior work fails to explore methods for efficiently fine-tuning VLAs for new tasks, a key component for adoption. Addressing these challenges, we introduce OpenVLA, a 7B-parameter open-source VLA trained on a diverse collection of 970k real-world robot demonstrations. OpenVLA builds on a Llama 2 language model combined with a visual encoder that fuses pretrained features from DINOv2 and SigLIP. As a product of the added data diversity and new model components, OpenVLA demonstrates strong results for generalist manipulation, outperforming closed models such as RT-2-X (55B) by 16.5% in absolute task success rate across 29 tasks and multiple robot embodiments, with 7x fewer parameters. We further show that we can effectively fine-tune OpenVLA for new settings, with especially

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